

ADR 0108 — A listener declares what it can honour, and a refusal outranks reassurance

- **Status:** Accepted — implemented and tested
- **Date:** 2026-09-02
- **Issue:** #589
- **Extends:** [ADR 0005](#), [ADR 0002](#)
- **Supersedes / superseded by:** —

Context

Git Vista deliberately builds two different HTTP routers. The loopback listener gets the complete route table. ADR 0005's LAN listener gets a structurally read-only table: its write, repository-selection, plan, preview, and write-outcome routes are not gated at runtime; they are never registered.

That boundary is correct. The live defect was on the other side of it:

listener	POST /api/select
127.0.0.1	401 – route exists; caller is unauthenticated
LAN address	405 – route does not exist; static fallback answered

The browser nevertheless rendered every repository as an actionable button. A click opened the next screen, `POST /api/select` received an ordinary 405, and the user ended up looking at a not-connected state. The console was empty because `fetch` rejects only for transport failures; an HTTP 405 is a fulfilled response.

The existing `SessionInfo.via_lan` flag did not solve the contract problem. It describes how a session was established. The route builder's `full_routes` value describes what the listener can honour. They happen to correspond in the two current constructors, but deriving a capability from network topology or a session fact gives the client a second source of truth. It also leaves the decision in `picker.rs`, which is compiled only for wasm32 and executed by no test runner in this repository.

A second live reproduction exposed the same false-confidence shape outside the picker. A 405 from `POST /api/plan` entered the preview panel's generic failure arm, followed by:

the operation ... is unchanged and still available

The response had established the opposite for this listener. A capability refusal may not be followed by reassurance that outruns it.

Decision

1. The listener declares a closed route-capability profile

Every listener response carries:

X-Git-Vista-Listener-Profile: full

or:

X-Git-Vista-Listener-Profile: read-only

`ListenerProfile` and `LISTENER_PROFILE_HEADER` live in the pure `git-vista-protocol` crate. Unknown or missing values never default to `full`. The server derives the value with

`ListenerProfile::from_write_routes(full_routes)` inside the same function whose `if full_routes` branch registers the complete table. It does not inspect Host, peer address, or `SessionState::via_lan` to reconstruct the answer.

The header is applied both to the API router and to the assembled app. The second placement matters: an absent LAN write route falls through to the static service, so its 405 is outside the API router and must still carry the listener's declaration.

This is protocol v11. A v10 client ignores the declaration and can reproduce the live-looking-button defect against a v11 server. That is semantic contract skew even though the catalog JSON itself did not change, so the compatibility window moves whole instead of allowing the old client through negotiation.

2. Catalog data and the declaration become one picker input

`fetch_catalog` reads the profile header before consuming the catalog body and returns one `RepositoryCatalog { repositories, listener_profile }`. The picker therefore cannot render repository data first and decide its action state later from an unrelated session store. If the declaration is absent or unknown, the catalog fetch fails visibly and no repository control is constructed.

The pure, host-tested `listener_policy` module maps the profile to one of two closed answers:

```
full      -> RepositorySelection::Offered
read-only -> RepositorySelection::Unavailable { visible notice }
```

The wasm-only view arranges that answer:

- a full-profile row is an enabled button;
- a read-only row is disabled and visibly says “Read-only LAN view — open the loopback link to switch repositories”;
- the action-shaped repository map is not offered for a read-only profile;
- Delete, Clone URL, and Rescan remain absent there, now from the declaration rather than from `via_lan` as a proxy.

The catalog still lists every repository on the LAN view. Read-only means the information is useful but switching the server-side session selection is not an available action.

3. A 405 is classified at both write funnels

Declaration is the honest surface, not a substitute for handling the answer. The shared retrying write transport inspects every completed response before returning it to an endpoint. Status 405 is converted to a visible refusal that names the route, the profile declared on that response, and the fact that the operation is unavailable there. A `read-only` declaration carries the loopback remedy. A `full` or missing/unknown declaration reports the mismatch and asks for a reload without falsely claiming that switching listeners will fix it.

Three read-shaped POSTs (`/api/plan`, `/api/preview`, and explicit diff) do not use that retrying transport. They already converge on `user_facing_error`, so the same pure classifier is applied there. A route-less fallback response has no API request id; that case is logged explicitly instead of preserving the empty console that hid #589.

Other HTTP statuses retain the server's structured message. In particular, 401 is still authentication, 403 is still an authorization or repository-mode refusal, and 404 is not silently rewritten as a listener-profile claim.

4. Preview request failure copy makes no availability claim

A successfully computed `PreviewOutcome` remains advisory and may keep the existing explanation that the preview does not gate execution. A failed request is a different state. Pending copy now

says only that the client is checking whether the listener can provide a preview. Failed copy reports the failure reason and stops.

The copy comes from the host-tested policy module. Its `/api/plan` 405 test assembles the exact rendered failure line and asserts both that it says `unavailable` and that neither “still available” nor “ready either way” can appear. The DOM module contains no second sentence that can contradict it.

Why both halves are required

The declaration and the 405 classifier close different failures:

```
profile declaration -> prevents a control that can only fail from looking live
405 classification  -> makes any stale, missed, or future control fail visibly
```

Declaration alone fixes the picker but leaves every other omitted route able to fail as an ordinary, easily flattened response. Classification alone improves the click's aftermath while preserving a button whose only possible outcome is refusal. Neither is an honest interface by itself.

Alternatives considered

Declare the profile only

Rejected. It prevents this picker click but provides no defence for another control accidentally wired to any of the many routes under `full_routes`. The next omission would again depend on every caller remembering that HTTP errors do not reject `fetch`.

Handle 405 only

Rejected. The user would receive a better message only after pressing a button that the client already knew could never work. A control is a claim that an action is available; error handling cannot make that initial claim honest.

Continue deriving capability from `SessionInfo.via_lan`

Rejected. Transport provenance and route capability are different facts. The router already owns the authoritative switch, and duplicating its meaning in a session mapping lets the declaration and the table drift. It also leaves the picker decision hidden behind `cfg(target_arch = "wasm32")`.

Put `profile` inside the catalog JSON or add a capabilities endpoint

Rejected in favour of the response header. A listener profile applies to the whole route table, not to repository records, and an extra endpoint creates another request that must finish before the catalog is safe to render. The header makes the answer atomic with the catalog response and is also present on the absent-route fallback that most needs to explain itself.

Send a list of every route or every UI control

Rejected. A route list makes transport names the frontend's feature model and requires a hand-maintained mapping from URLs to controls. A list of controls reverses the coupling and makes the server know DOM concepts. The two closed profiles match the router's one structural branch; the 405 defence covers future omissions without duplicating that route list.

Register `/api/select` (or `/api/plan`) on the LAN router and refuse inside

Rejected. ADR 0005's security property is structural absence: a later mode-gate regression cannot reopen a handler that was never put in the router. The LAN router is right and remains unchanged.

Disable rows without visible copy, or explain with a tooltip

Rejected. A disabled unexplained row is another apparent no-op, and a tooltip is not a disclosure on touch or assistive technology. The remedy is visible inside every disabled row.

Treat 405 as an expired session

Rejected. Loopback's 401 and LAN's 405 are measured, different answers. Turning the latter into sign-out discards the correct machine fact and confidently invents an authentication failure.

Verification

The decision lives outside wasm-only modules. Native tests execute the profile mapping, picker policy, 405 classifier, and preview failure copy; wasm checking then verifies that the DOM and `gloo-net` glue consume those answers.

Invariant	Executable check
full_routes and the response declaration use one closed mapping	listener::tests::the_route_switch_and_header_declaration_are_one_mapping
Unknown declarations fail closed	listener::tests::an_unknown_or_weakened_declaration_never_defaults_to_full
Real loopback/LAN route tables declare full / read-only	tests::the_loopback_router_still_has_write_routes_registered ; tests::the_lan_router_has_no_write_routes
The measured 401/405 pair and the header cannot disagree	tests::select_route_presence_and_listener_declaration_cannot_disagree
A read-only picker policy is disabled and carries visible remedy text	listener_policy::tests::a_read_only_profile_never_offers_repository_selection_and_s listener_policy::tests::the_picker_wires_the_policy_to_both_action_and_explanation
Only 405 becomes this capability refusal, and both POST funnels ask that classifier	listener_policy::tests::only_method_not_allowed_becomes_a_listener_capability_refus listener_policy::tests::both_wasm_post_funnels_consult_the_host_tested_405_classifi
/api/plan 405 cannot be followed by availability reassurance in either policy or DOM glue	listener_policy::tests::a_plan_405_can_never_be_followed_by_an_availability_promise listener_policy::tests::the_failed_preview_arm_cannot_append_reassurance_behind_the

Local verification before mutation proof:

- cargo test -p git-vista : **810 passed, 2 ignored** in the real git-vista-ui binary; the 0-test lib.rs target is reported separately and was not used as evidence.
- cargo test -p git-vista-protocol : **229 passed** across unit/integration targets, with one ignored doctest.
- cargo test -p git-vista-server : **1,138 passed, 6 ignored** across its binary/integration targets; the three named router/profile tests pass against real Axum routers.
- cargo check -p git-vista --target wasm32-unknown-unknown passes.
- cargo clippy --workspace --all-targets -- -D warnings and the corresponding wasm32 frontend clippy pass.
- trunk build passes and produces the real wasm bundle.

- `cargo fmt --all -- --check` and `git diff --check` pass.

Mutation matrix

`failure-atlas mutation_check` ran against committed HEAD `600ee784`. Every unmutated baseline was green, every edit applied exactly once inside the atlas's contained clone, and every mutated leg reached and failed an assertion; no compiler failure is counted as a catch. The atlas reported the source tree as dirty solely because of the user's pre-existing untracked `.grok/` directory. It cloned the recorded HEAD, and no tracked implementation or test change was pending.

Invariant	Remove mechanism	Weaken / misroute	Result
Read-only profile does not offer repository selection and carries the remedy	map <code>ReadOnly</code> to <code>Offered</code>	replace the row's <code>disabled=!can_select</code> with <code>disabled=false</code>	caught/caught — records 155-156; the policy assertion and wasm-seam assertion fail independently
A 405 becomes a visible listener refusal, while other statuses retain their meaning	make the classifier always return false	widen it to every status ≥ 400	caught/caught — records 157-158; 405 disappears in the first and ordinary statuses are misclassified in the second
<code>/api/plan</code> 405 never produces availability reassurance	append “still available” in the host-tested failed-preview formatter	append the same promise directly in the wasm failed arm	caught/caught — records 159-160; the rendered-line assertion and failed-arm seam assertion each fail
Router profile declaration matches the route set	map <code>full_routes == false</code> to <code>Full</code>	make only the assembled app stamp <code>Full</code> over its read-only API router	caught/caught — records 161-162; the closed mapping and live 405/header assertions fail respectively

The assertion that detects the original silent path is the 405 classifier test: removing classification makes it receive `None` rather than a listener-specific refusal. The independent picker assertions still pass in that mutation, which is why both halves are pinned. Overall failure-atlas result: **8 caught, 0 survived, 0 baseline/build failures**.

Consequences

- The LAN picker remains useful as a catalog but no longer pretends it can switch the selected repository.
- An unexpected 405 from any ordinary write or direct plan/preview POST becomes visible and actionable instead of looking like navigation or lost auth.
- A `/api/plan` refusal can no longer share a panel with words claiming the operation remains available.
- Adding another listener profile is a closed-contract change: the protocol enum, router mapping, client policy, tests, and compatibility decision must be revisited together.
- The wasm-only modules keep DOM/fetch glue; the decisions that can go wrong are exercised by the host test runner.

Signed: max · 2026-09-02